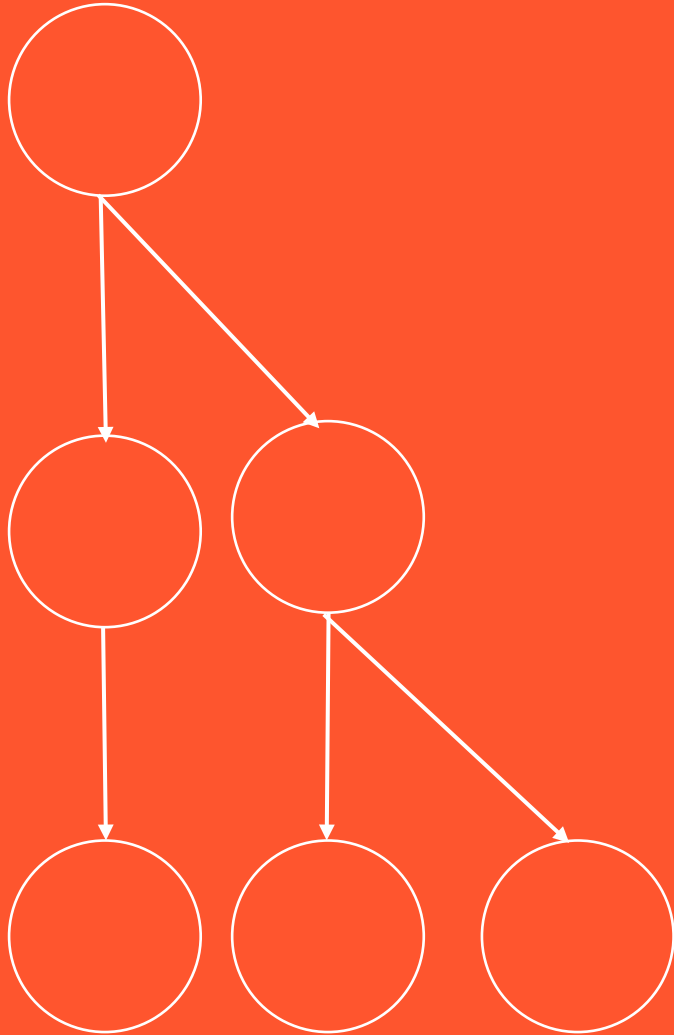


BCOG 100

Memory, Simulation, & Generative Models

Lecture 1: The Evolution of the Cortex



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 9-1

Password: memory

Question: What is memory? (according to the article you read)

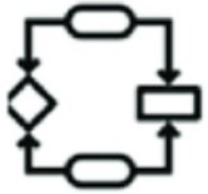


The Evolution of the Cortex



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

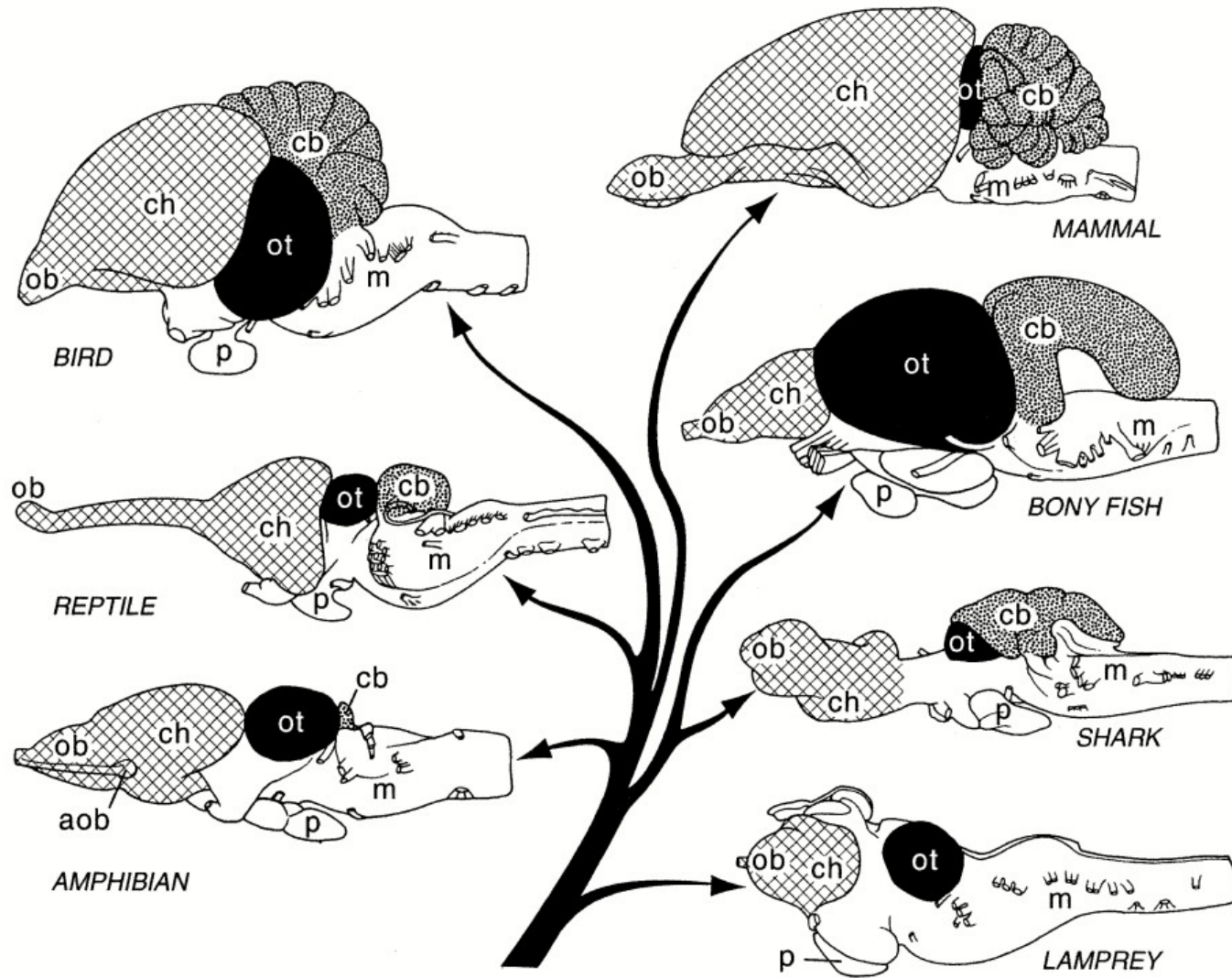
Last Time:

Spatial cognition as the basis for memory

Today:

- Prediction takes center stage
- brain evolved to anticipate the world farther into the future.
- Simulation—imagining before acting—was the next great leap.

The Evolution of the Cortex



The Neural Dark Ages

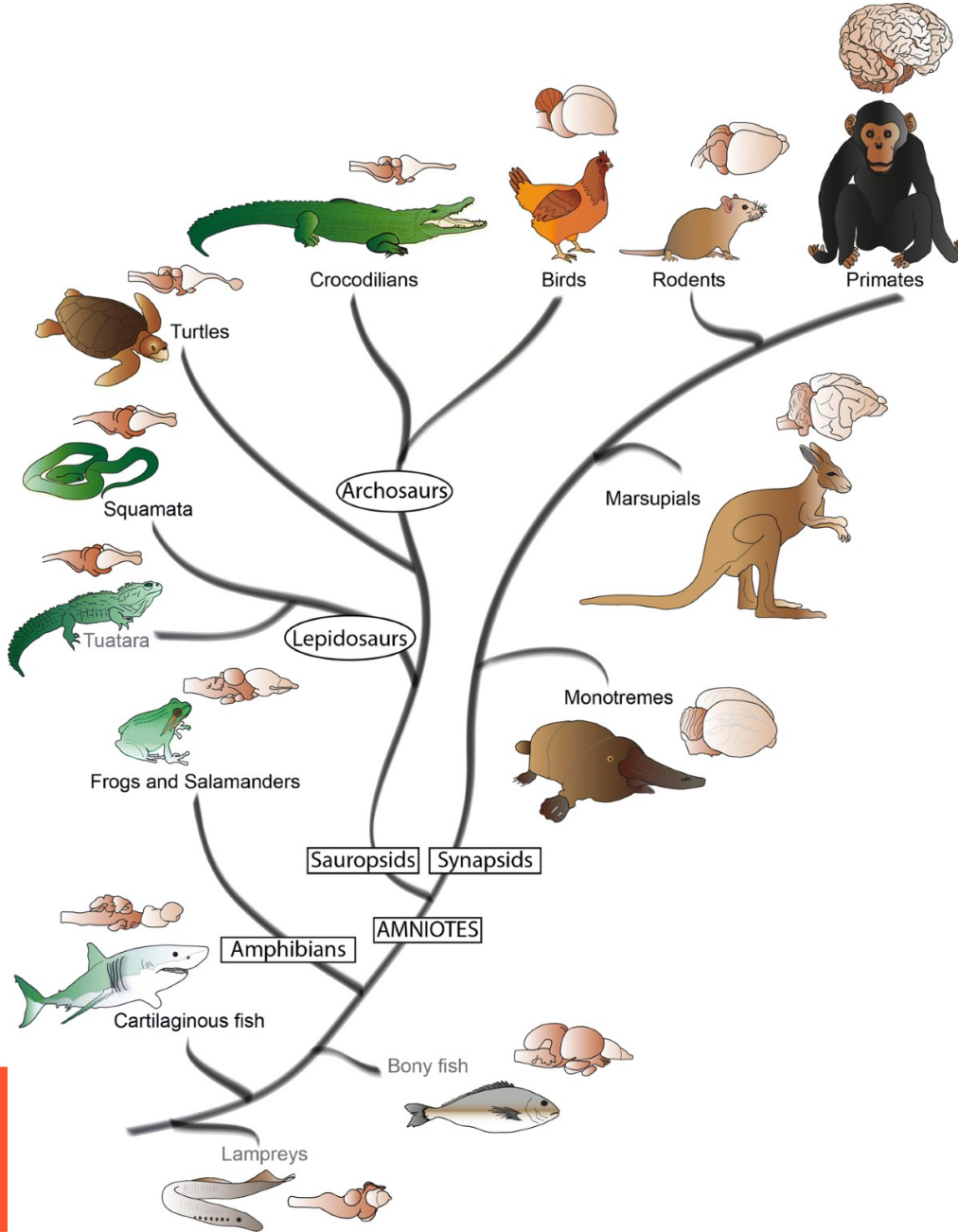
- For hundreds of millions of years, vertebrate brains barely changed.
- Evolution focused on bodies, not minds.
- The cortex lay dormant—until mammals rewired it for imagination.

The Evolution of the Cortex

Event	Approx. age (Mya)
End-Ediacaran biotic turnover	~550
Cambrian Explosion	~540
Origin of vertebrates (jawless fish)	~520–500
End-Ordovician mass extinction	~444
Evolution of jawed fish	~430–420
Late Devonian mass extinction	~372–359
Evolution of land vertebrates (early tetrapods)	~375–360
End-Permian mass extinction	~252
Evolution of mammals (mammal lineage diverges from reptilian lineage)	~220–200
End-Triassic mass extinction	~201
End-Cretaceous mass extinction	~66
Radiation of mammals	~60



The Evolution of the Cortex



Life on Land and the Rise of the Amniotes

- Leaving the water demanded new kinds of control.
- Land animals needed brains that could plan movement, not just react.
- The amniote brain laid the groundwork for flexible behavior.

The Evolution of the Cortex

Warm-Bloodedness and Its Cognitive Costs

Feature	Endothermy (Warm-Blooded)	Ectothermy (Cold-Blooded)
Activity Level	Can remain active in cold or dark	Activity limited by ambient warmth
Reaction & Processing Speed	Fast neural conduction; rapid, sustained movement	Slower reactions in cool environments
Ecological Advantage	Expanded temporal niche (e.g., nocturnal, cold-weather activity)	High efficiency in stable, warm climates
Energy Demand	Extremely high	Low, energy-efficient
Growth & Reproduction	Requires continuous energy intake; fewer offspring	Can survive long periods without food
Cognitive Potential	Supports sustained neural activity	Limits prolonged complex cognition
Evolutionary Trade-Off	Speed and flexibility at high cost	Efficiency and simplicity at low cost

The Evolution of the Cortex

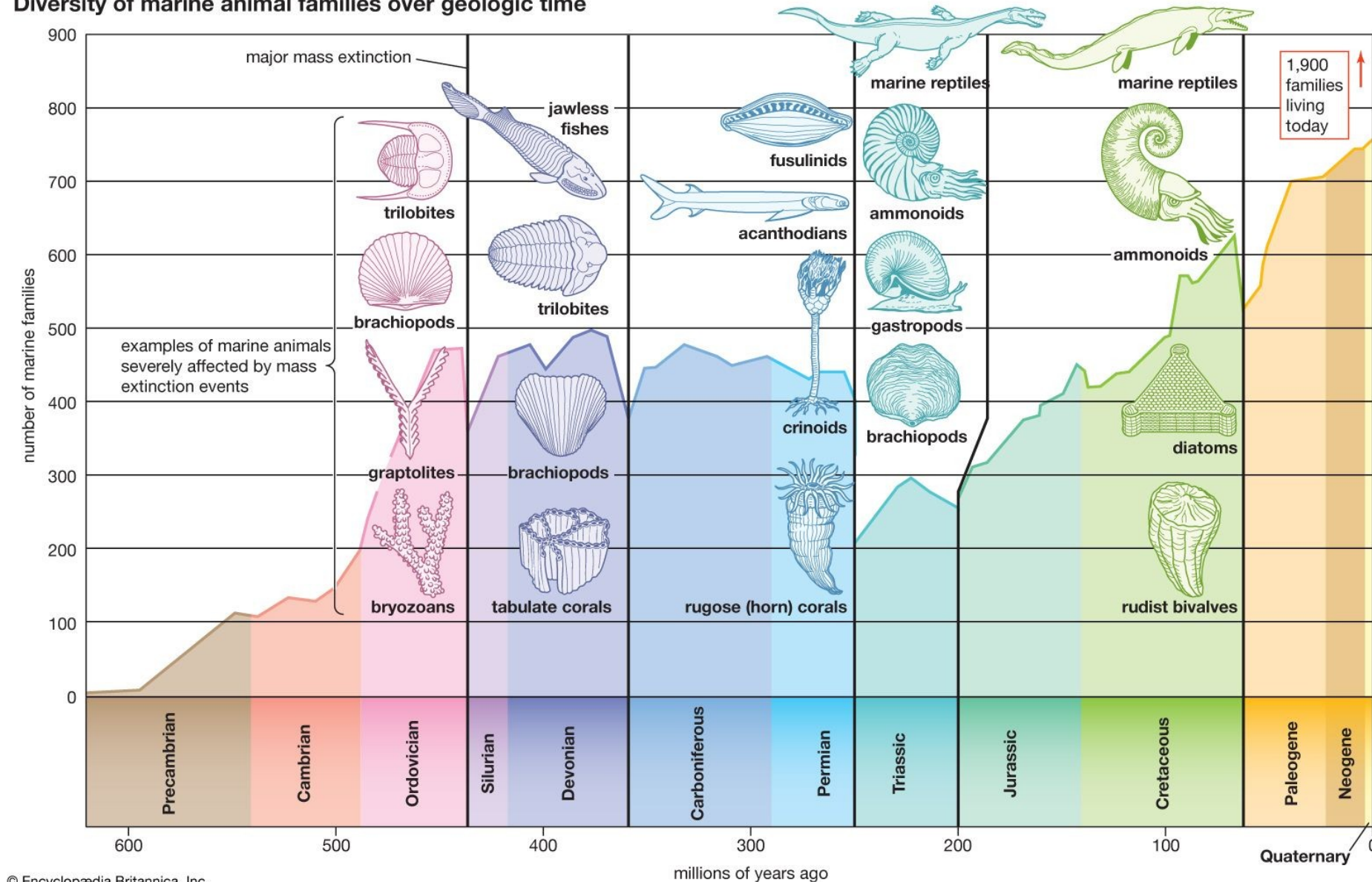


The Therapsid Gamble – Energy for Flexibility

- Early mammal ancestors traded efficiency for adaptability.
- They burned more energy to stay active and aware.
- Flexibility, not power, became the key to survival.

The Evolution of the Cortex

Diversity of marine animal families over geologic time



Surviving the Permian Extinction

- When most species vanished, flexibility became everything.
- Therapsids with adaptive minds endured catastrophe.
- Cognition was born in crisis.

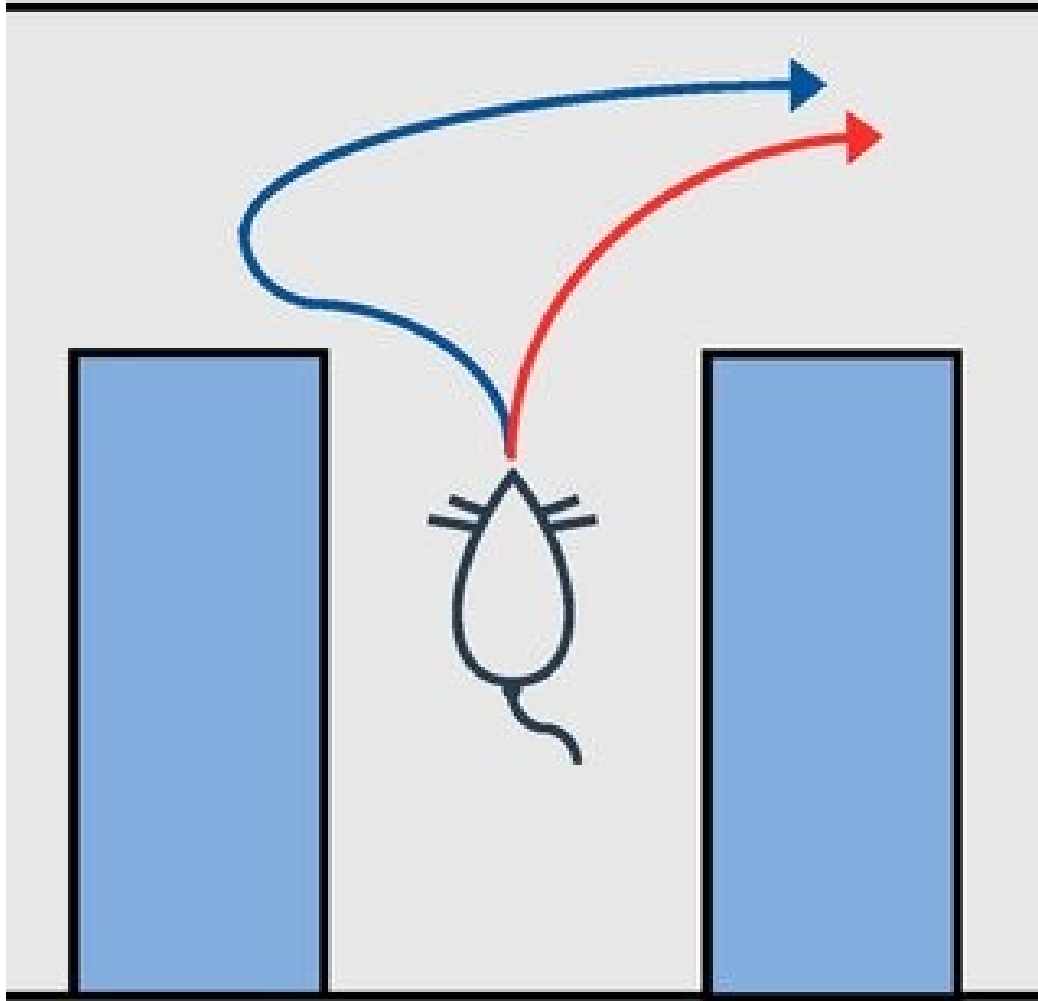
The Evolution of the Cortex



The Rise of Nocturnal Mammals

- In a dinosaur's world, mammals survived by night.
- Darkness favored quiet, cautious, imaginative brains.
- Planning replaced brute force.

The Evolution of the Cortex

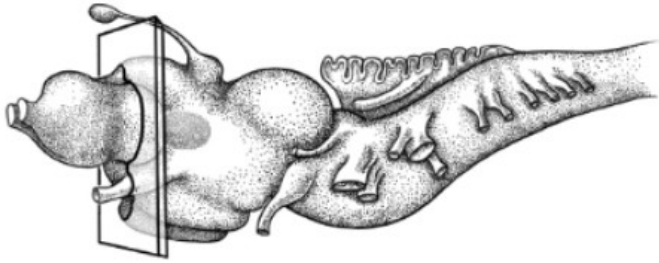


The Gift of the First Move – Planning Before Acting

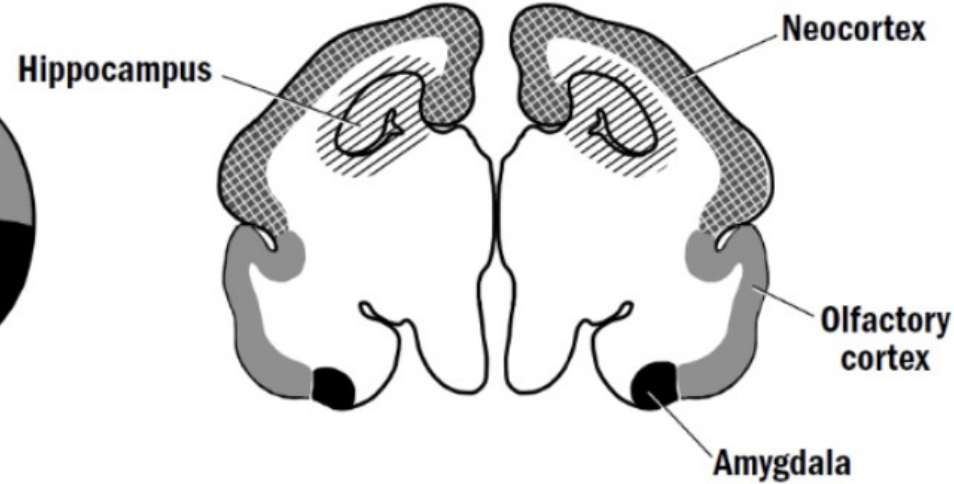
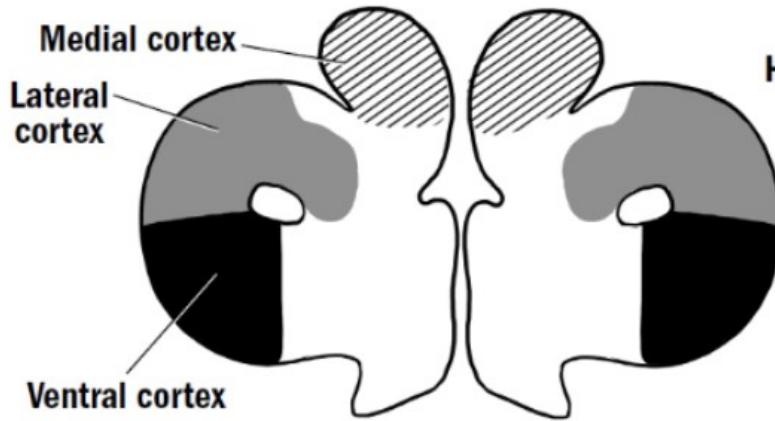
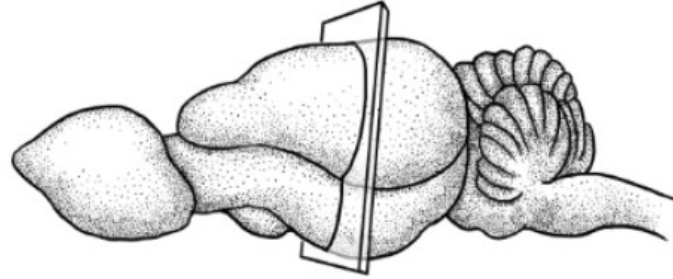
- Reptiles learned by doing.
- Mammals could also learn by imagining doing.
- Simulation let them choose the best action before committing to it.
- “Vicarious Trial and Error”

The Evolution of the Cortex

Early vertebrate cortex



Early mammal cortex



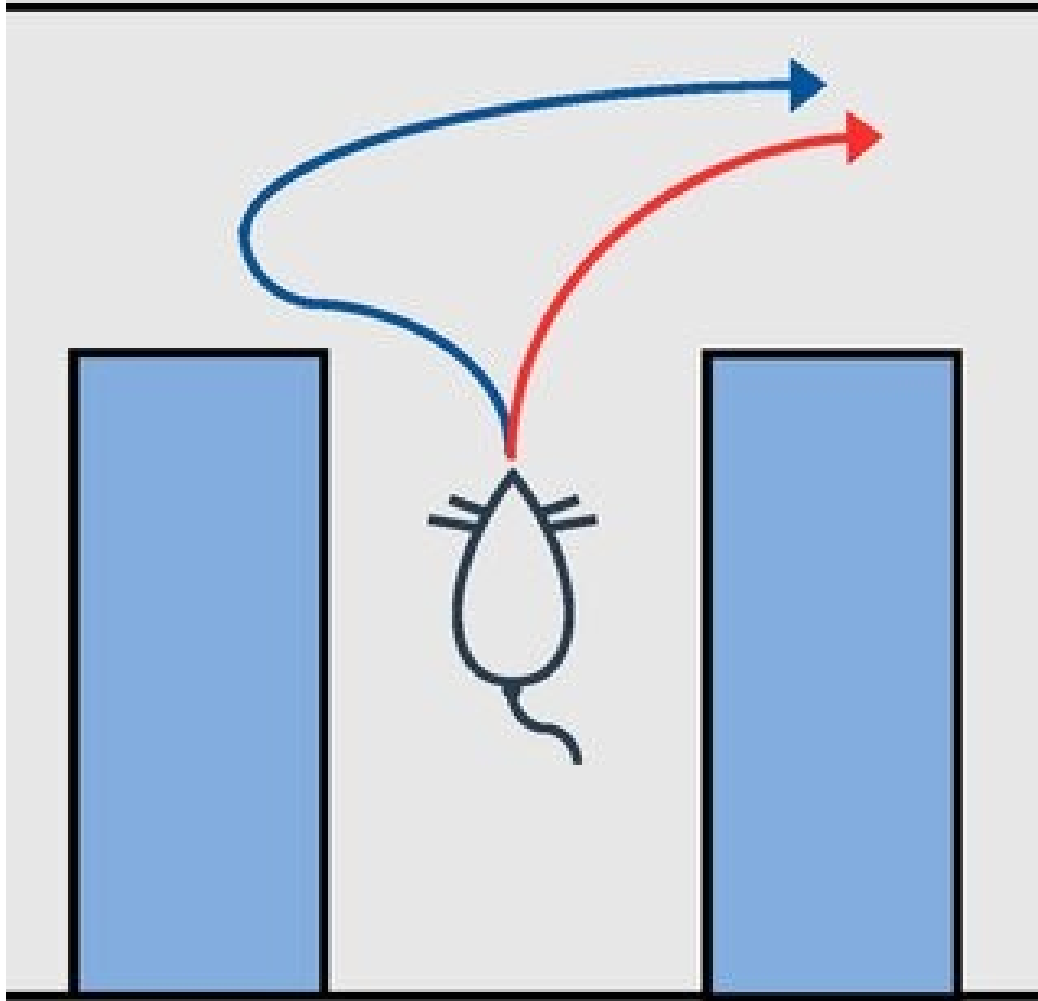
The Birth of the Neocortex

- A new cortical layer changed everything.
- The neocortex added flexible, general-purpose computation.
- It allowed animals to *model* the world, not just sense it.

Figure 10.3: How the cortex changed in the transition from early vertebrates to early mammals

Original art by Rebecca Gelernter

The Evolution of the Cortex



From Learning by Doing → Learning by Imagining

- Reptiles learned through experience.
- Mammals began to learn from *simulated* experience.
- The brain could now run experiments inside itself.

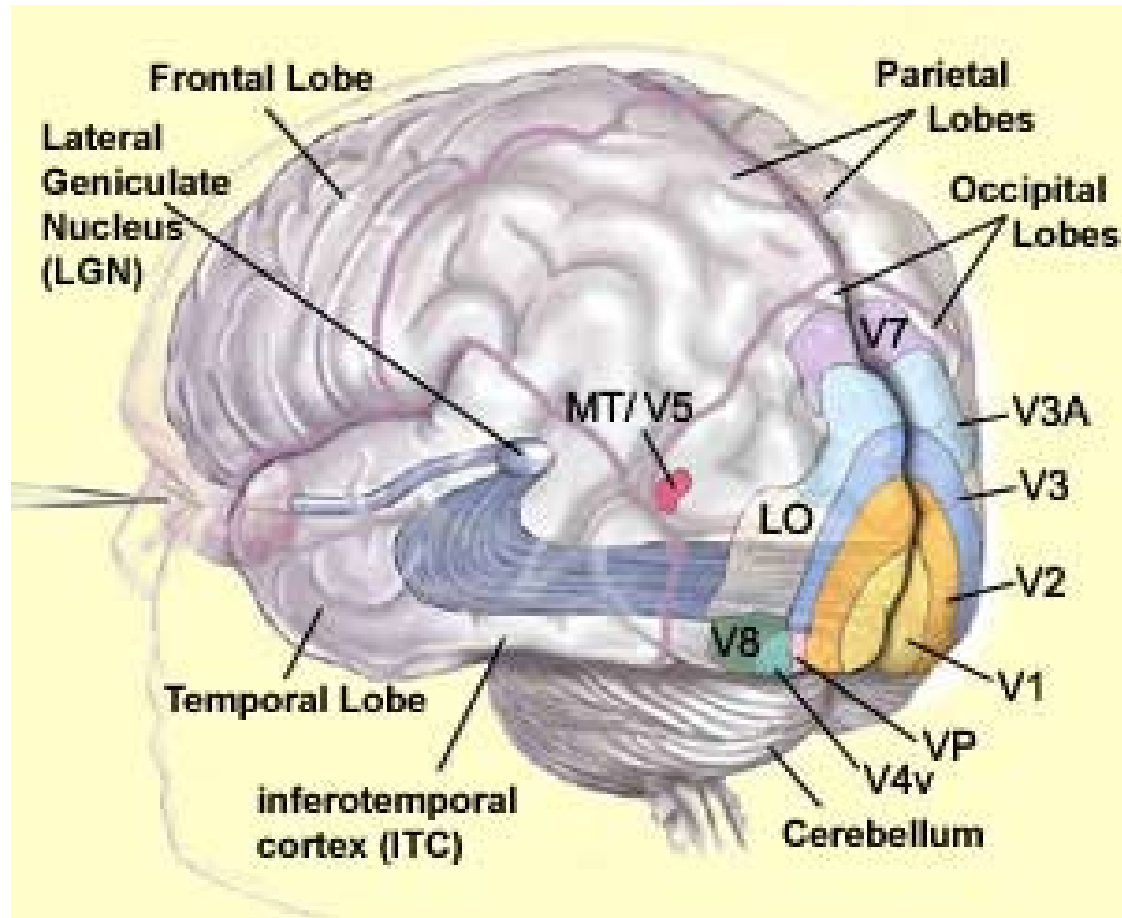
The Evolution of the Cortex



Why Simulation Was an Evolutionary Breakthrough

- Simulation solved the limits of trial-and-error learning.
- It made learning faster, safer, and more flexible.
- Thinking became an evolutionary shortcut.

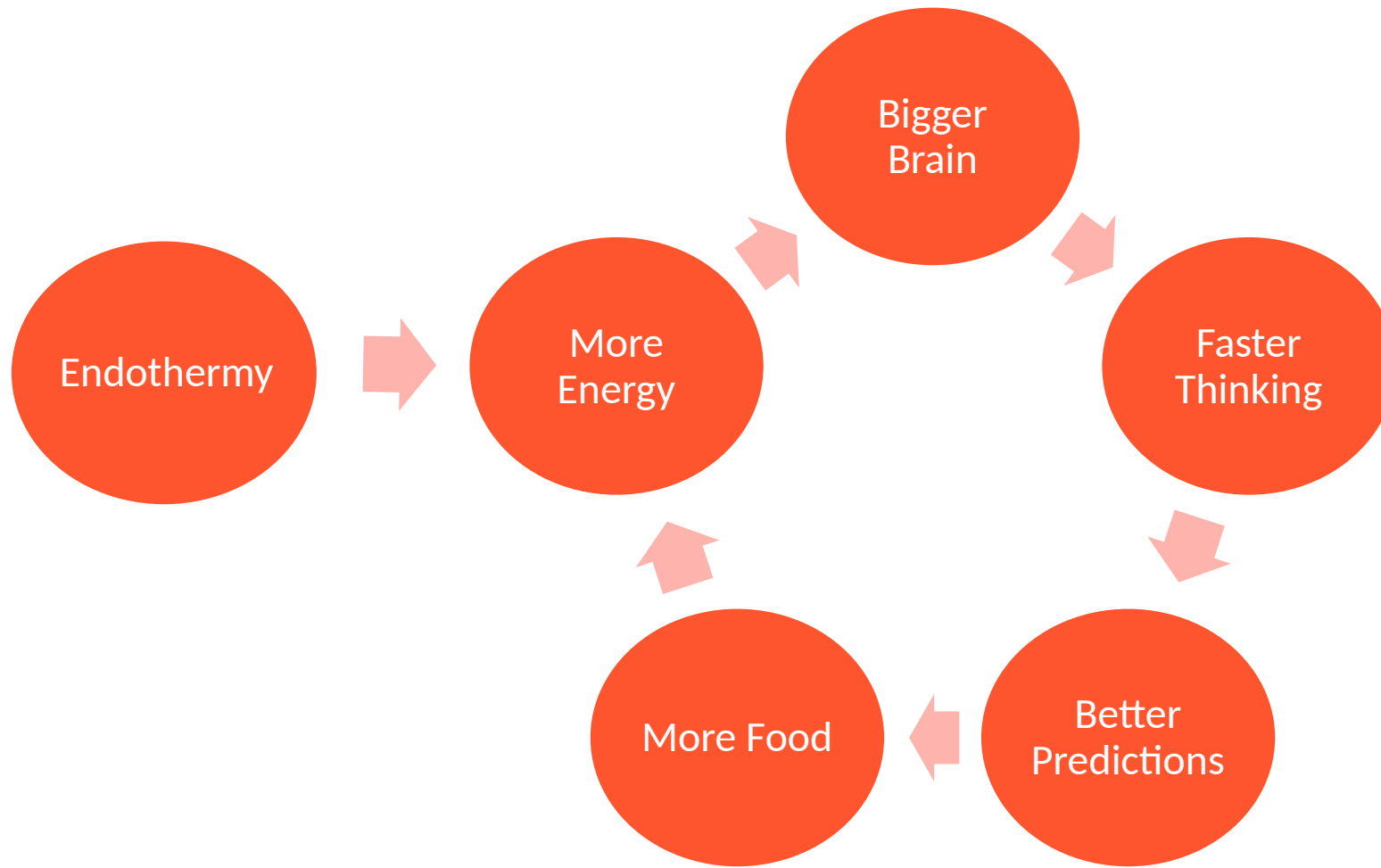
The Evolution of the Cortex



Vision as a Gateway to Simulation

- Many senses can support imagination.
- But vision offered a natural foundation for internal modeling.
- Its spatial structure made prediction possible.

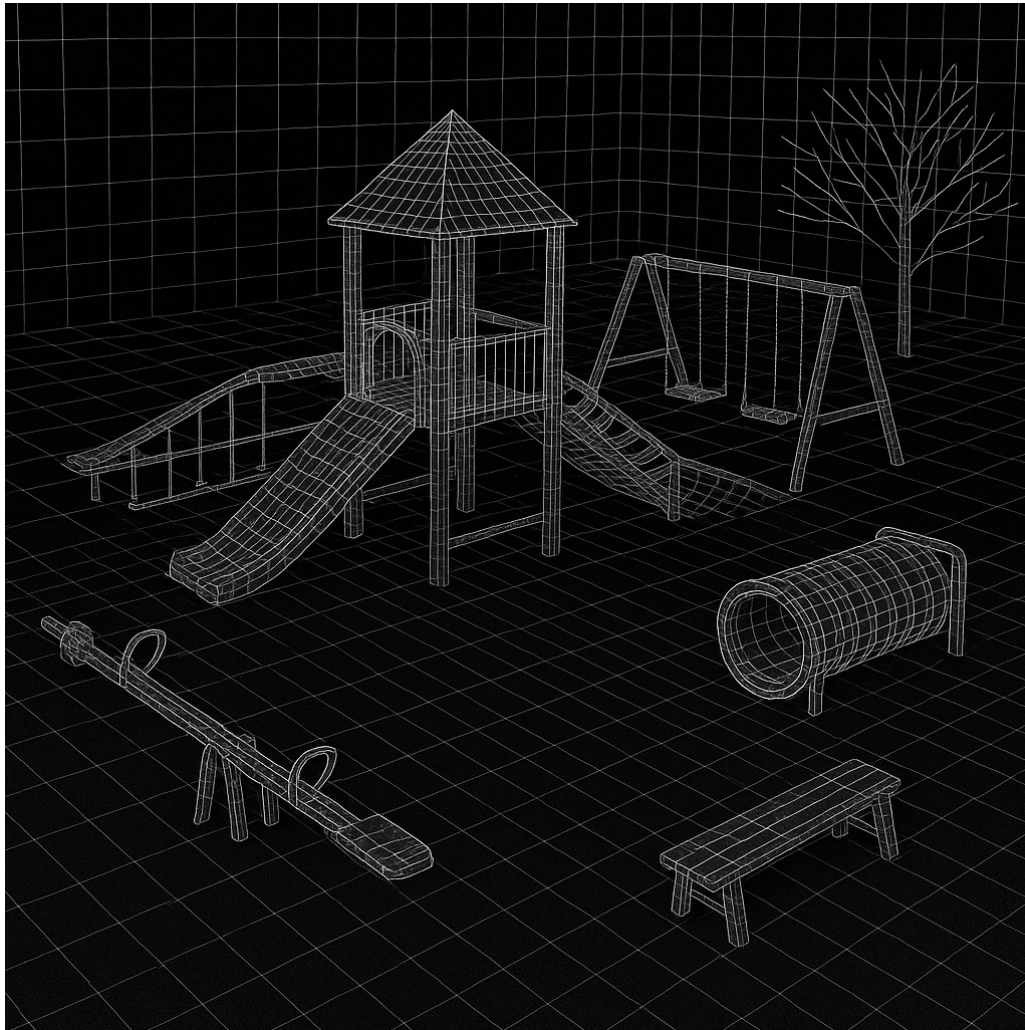
The Evolution of the Cortex



Warm Brains = Fast Computation

- Warm-bloodedness sped up neural processing.
- Fast neurons made predictive thinking possible.
- Energy fueled foresight.

The Evolution of the Cortex



The Third Great Predictive Leap: World Models

- Reflexes predicted milliseconds.
- Reinforcement predicted seconds.
- World models let brains predict minutes, hours, and beyond.

The Evolution of the Cortex

SIMILAR FUNCTIONS, DIFFERENT PLACEMENTS

Birds' brains are equipped with sensory and cognitive processing centers roughly equivalent to those in primates. But their placement can differ. The nidopallium caudolaterale (NCL) at the back of the brain, for instance, serves as an integrating hub for all of the animals' sensory, limbic and motor systems—similar to the prefrontal cortex in primates.

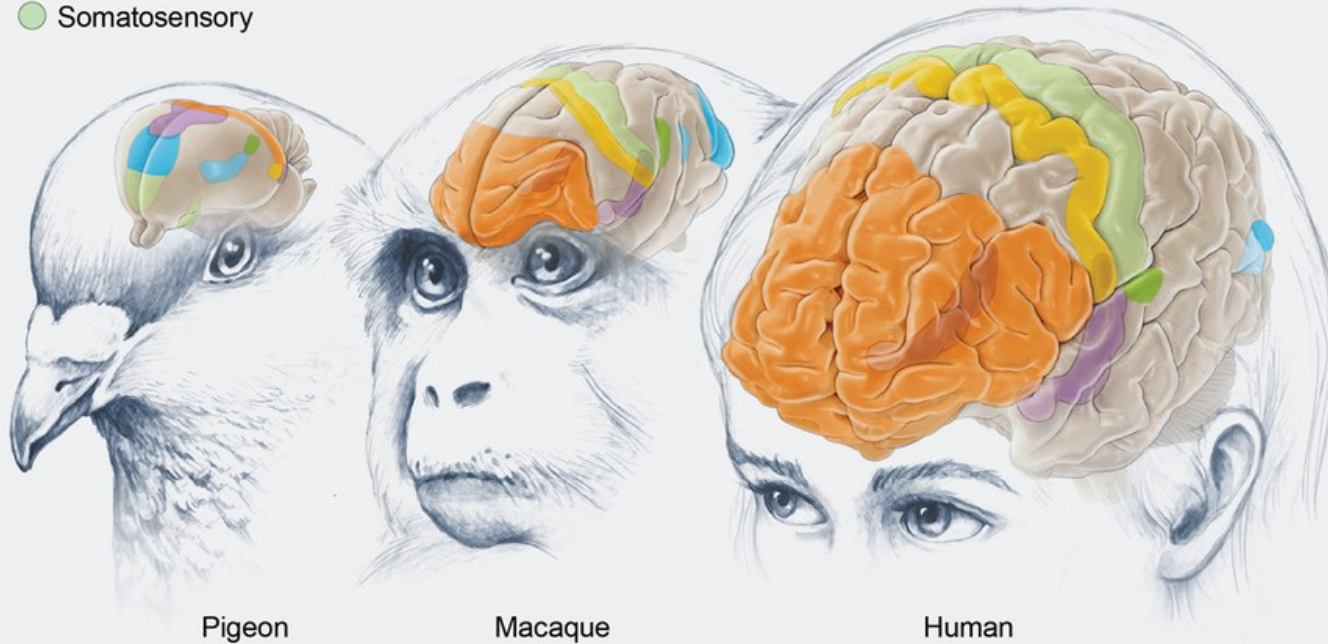
Sensory System

- Visual pathway 1
- Visual pathway 2
- Auditory
- Somatosensory

Emotional/Memory System

- Hippocampus
- Amygdala

- Prefrontal/NCL
- Motor



Pigeon

Macaque

Human

Birds – A Parallel Evolution of Warm-Blooded Thought

Endothermy evolved twice—and so did foresight.

Birds built their own simulation machinery.

Different anatomy, same computational solution.

The Evolution of the Cortex

World Models

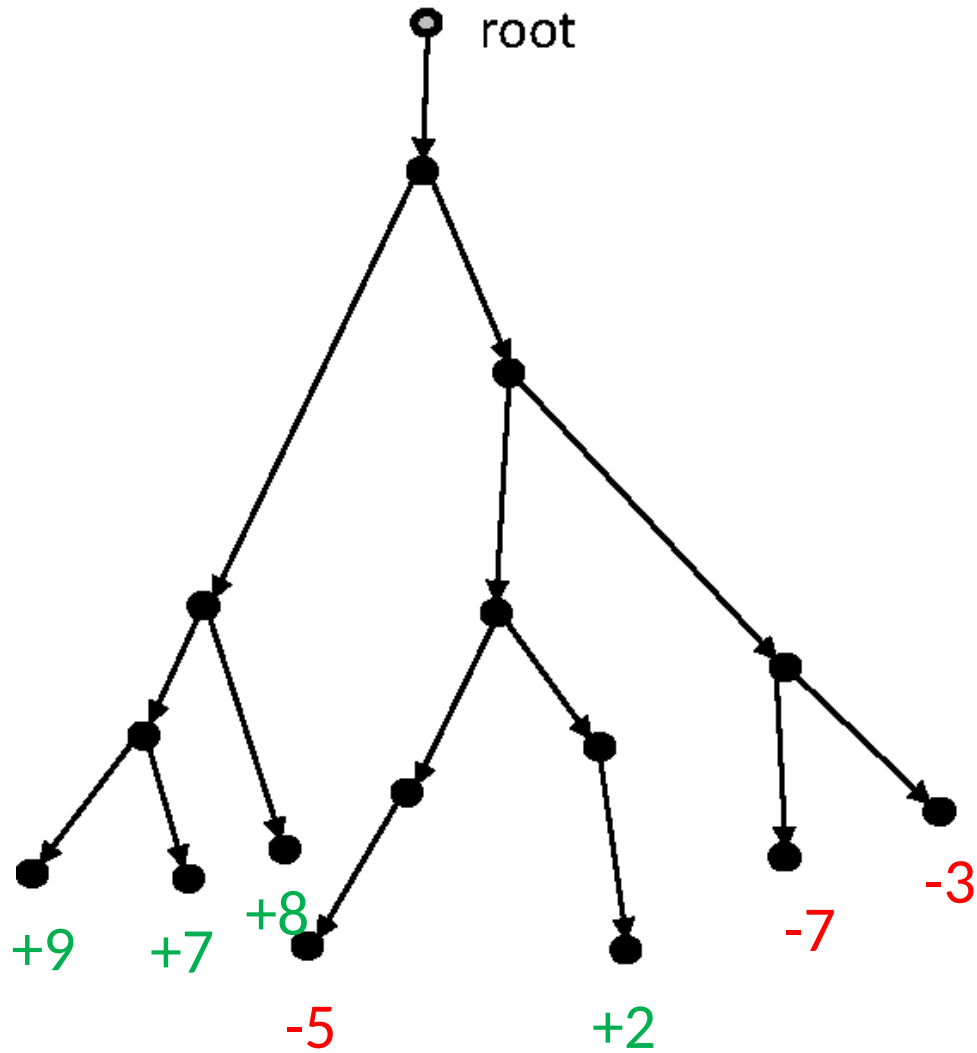
Hebbian &
Reinforcement
Learning

Reflex
Arcs

**Reflex → Reward →
World Prediction**

- Reflexes react.
- Reinforcement learns from outcomes.
- World models imagine outcomes before they happen.

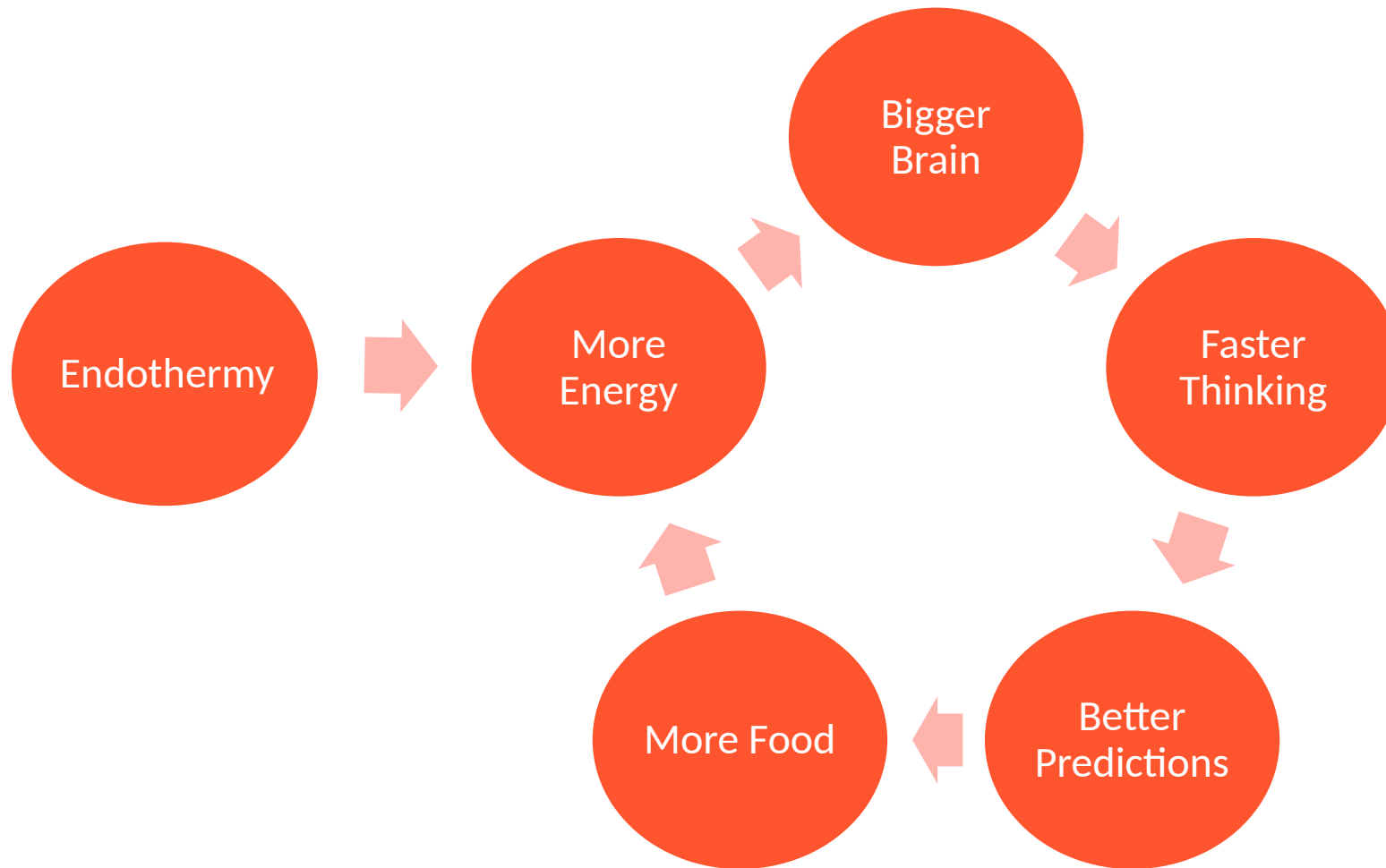
The Evolution of the Cortex



Functional Payoffs – Planning, Exploration, and Learning from Absence

- Simulation made action smarter, not faster.
- Mammals could plan routes, test ideas, and learn from what *didn't* happen.
- Imagination expanded the meaning of experience.

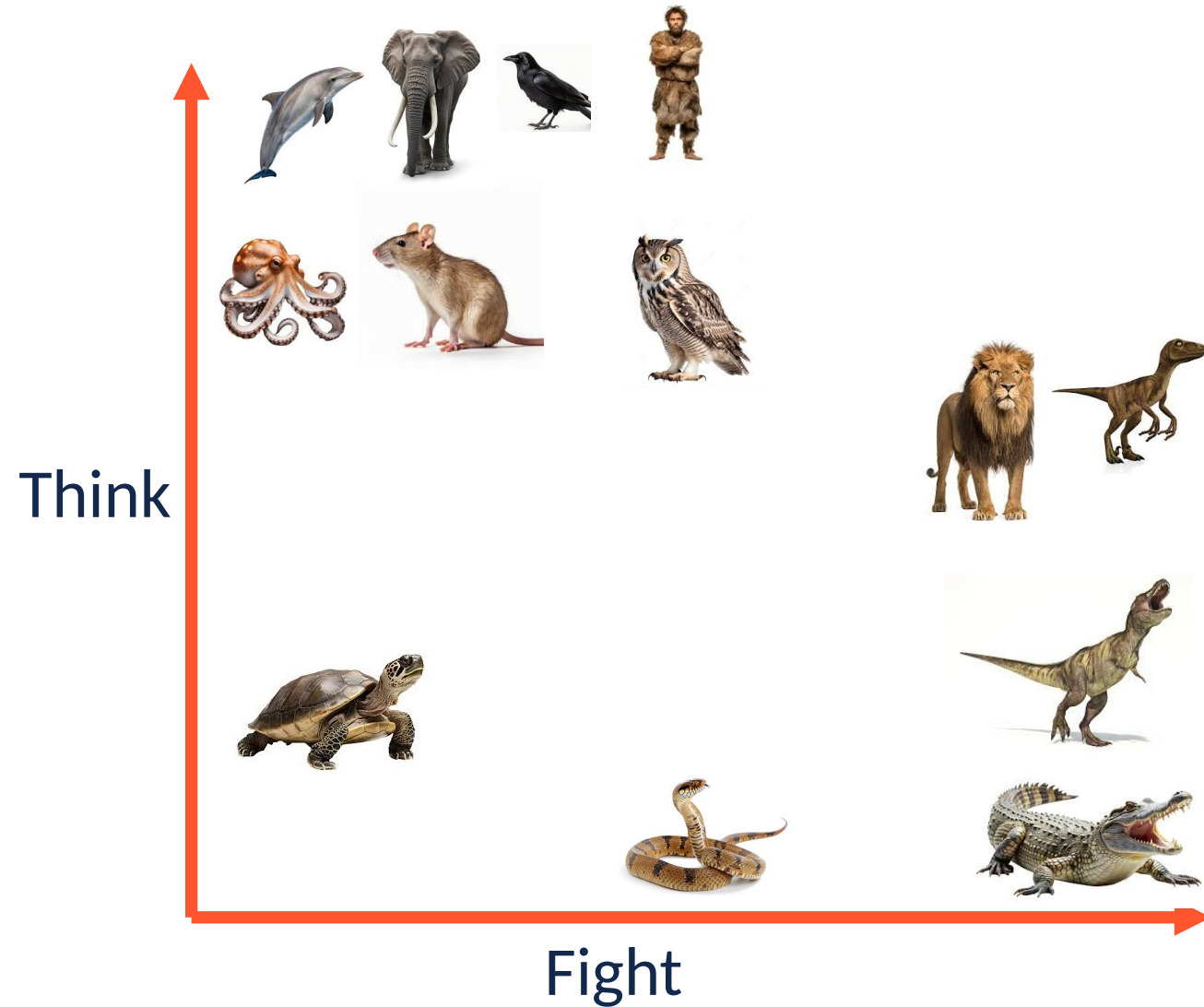
The Evolution of the Cortex



Energy Trade-Offs and Metabolic Costs

- Prediction is expensive.
- Brains burn enormous energy to imagine futures.
- But foresight saves more energy than it spends.

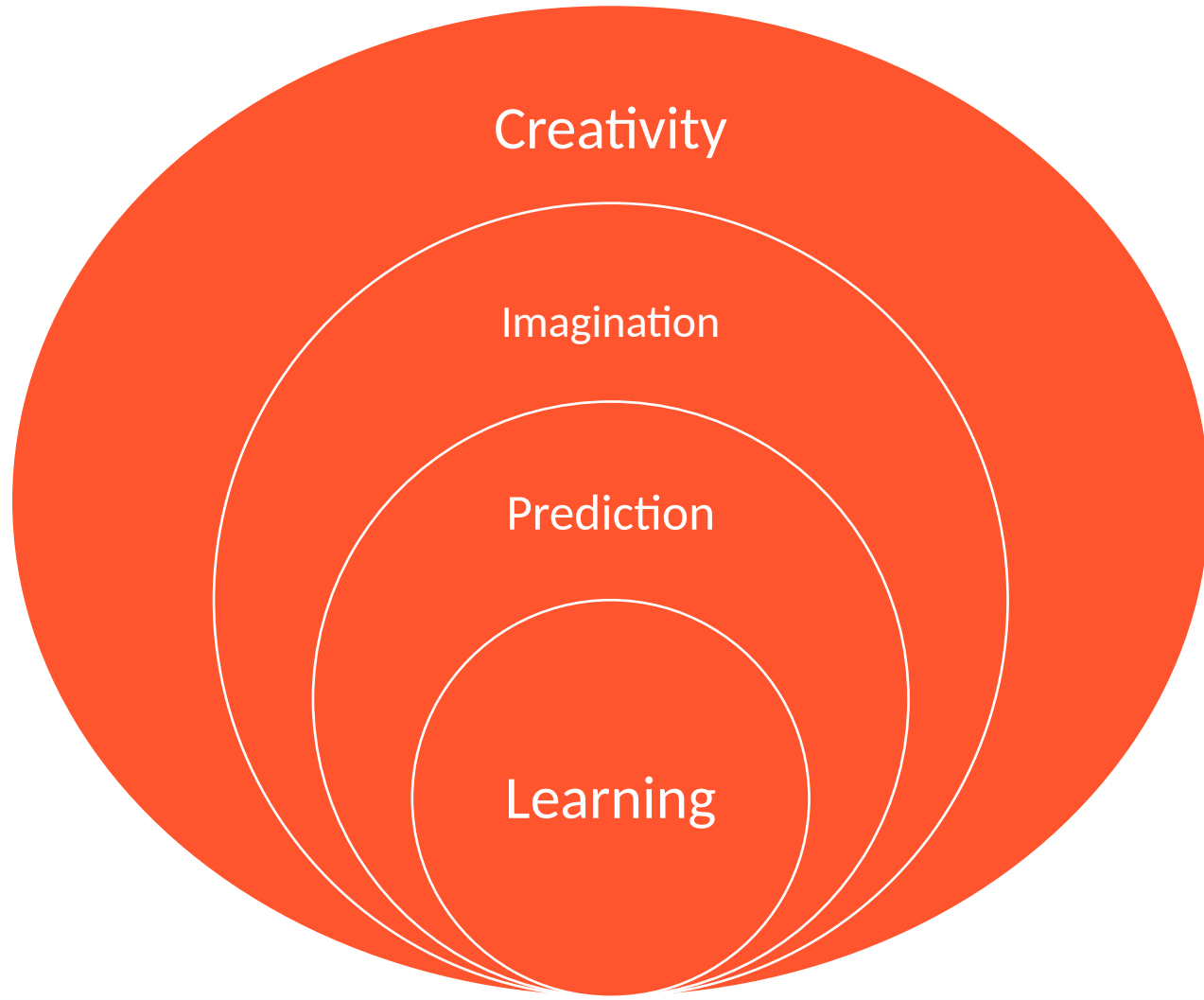
The Evolution of the Cortex



Simulation as a New Cognitive Niche

- Mammals didn't outfight their rivals—they outthought them.
- Simulation became a new ecological strategy.
- Thinking was a way of surviving.

The Evolution of the Cortex



From Survival to Imagination – The Seed of Mind

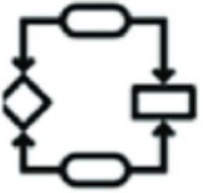
- Imagination began as a survival tool.
- But once brains could simulate the world, they could imagine *anything*.
- Survival gave rise to creativity.

The Evolution of the Cortex



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap

- Simulation let mammals act without trial and error.
- It turned experience into imagination and foresight.

Up Next

How the brain *implements* this miracle of prediction.